

Auteur: Viki Lauvrys  
Studierichting: Informatica  
Oefeningenreeks 4A nr. 30.3

$$\int x^3 \sqrt{1-x^2} dx$$

$$\text{Stel } t = 1 - x^2 \Rightarrow dt = -2x dx$$

$$\Rightarrow x = \sqrt{1-t} \Rightarrow dx = \frac{dt}{-2x}$$

$$= \int (\sqrt{1-t})^3 \sqrt{t} \frac{dt}{-2(\sqrt{1-t})}$$

$$= \int \frac{(1-t)\sqrt{1-t}\sqrt{t}}{-2(\sqrt{1-t})} dt$$

$$= \int \frac{(1-t)\sqrt{t}}{-2} dt$$

$$= -\frac{1}{2} \int (\sqrt{t} - \sqrt{t^3}) dt$$

$$= -\frac{1}{2} \int \left( t^{\frac{1}{2}} - t^{\frac{3}{2}} \right) dt$$

$$= \frac{-t^{\frac{3}{2}}}{\frac{3}{2}} + \frac{t^{\frac{5}{2}}}{\frac{5}{2}} + C$$

$$= \frac{-(1-x^2)^{\frac{3}{2}}}{\frac{3}{2}} + \frac{(1-x^2)^{\frac{5}{2}}}{\frac{5}{2}} + C$$

## References